

PRD — Multi-Tenant User Social Media Scheduler SaaS (n8n-Centric)

Executive Summary

Build a multi-tenant SaaS allowing users to connect Facebook + Instagram using Meta OAuth, upload media + captions, schedule posts, and publish automatically via n8n workflows. Stack: Next.js frontend, Supabase (Auth/Postgres/Storage), n8n as primary backend/orchestrator, Meta Graph API.

Goals & MVP Scope

MVP: signup/login, connect FB+IG, upload media, create/edit/cancel drafts, timezone scheduling, logs/history, retry mechanism, multi-account support. V2: recurring posts, analytics, AI captions, approval workflows.

Architecture

Frontend → Supabase Auth → n8n Webhooks/API → Supabase DB/Storage → Meta Graph API.
n8n owns API backend, OAuth callbacks, scheduler, retries, token refresh, logging.

Core Tables (Supabase)

users(id, email, created_at)
social_accounts(id,user_id,platform,page_id,ig_user_id,access_token,refresh_token,expires_at,status)
media_assets(id,user_id,file_url,file_type,size,created_at)
scheduled_posts(id,user_id,status,title,caption,schedule_at,timezone,target_platforms,created_at,updated_at)
post_media(post_id,media_id)
post_logs(id,post_id,workflow,status,error_message,response_payload,created_at)
workflow_runs(id,workflow_name,status,duration,created_at)

RLS Policies

Users can only SELECT/INSERT/UPDATE their own rows using auth.uid() mapping. Tokens encrypted at rest. Service role restricted to n8n backend tasks.

n8n Workflows

1. OAuth Connect Workflow — create Meta authorization URL.
2. OAuth Callback Workflow — exchange code for token, fetch account/page IDs, store encrypted tokens.
3. Upload Media Workflow — validate upload, store in Supabase Storage, persist metadata.
4. Create Post Workflow — validate payload, create draft/scheduled record.
5. Scheduler Cron Workflow — every minute query pending posts WHERE schedule_at <= now().
6. Publish Workflow — fetch media, choose platform adapter, call Meta API, update status.
7. Retry Workflow — exponential backoff (1m/5m/30m). Max 3 retries.
8. Token Refresh Workflow — refresh expiring tokens.
9. Logging Workflow — central structured logs.

API Contracts (n8n Webhooks)

POST /api/oauth/connect
GET /api/oauth/callback
POST /api/media/upload
POST /api/post/create
PUT /api/post/edit/:id
POST /api/post/cancel/:id
GET /api/posts
GET /api/logs

Meta Integration

Use Meta Graph API only. Required scopes: `pages_manage_posts`, `pages_read_engagement`, `instagram_basic`, `instagram_content_publish`, `business_management`.
Instagram requires Business/Creator account linked to Facebook Page.

Frontend Pages

Auth, Dashboard, Composer, Scheduled Posts, Drafts, Connected Accounts, History/Logs, Settings.

Security

Encrypted OAuth tokens, signed webhooks, server-side validation, upload MIME validation, rate limiting, audit logs, secrets via env vars.

Deployment

Dockerized deployment. Services: Next.js, n8n, Supabase. Use reverse proxy + HTTPS. Environment variables for Meta credentials and Supabase keys.

Claude Generation Instructions

Generate implementation artifacts in this order:

1. Supabase SQL schema + RLS.
2. n8n workflow JSON for each workflow.
3. Next.js frontend screens.
4. Meta API integration logic.
5. Docker compose deployment.

All outputs must be production-ready, typed, modular, and environment-variable driven.

Claude Master Prompt

You are a senior SaaS architect. Using this PRD, generate a complete implementation for a multi-user n8n-centric social media scheduler using Next.js, Supabase, Supabase Storage, Supabase Auth, Meta Graph API, and n8n as the primary backend. Produce: SQL schema, RLS policies, environment variables, n8n workflow JSON, webhook definitions, API payload schemas, React pages, deployment configs, retries, token refresh logic, structured logging, and production security defaults.